PROGRAM:

```
#include <stdio.h>
int main()
{
   int a, b, res, choice;
   printf("===== BITWISE OPERATIONS =====\n");
   printf("Enter the first number: ");
   scanf("%d", &a);
   printf("Enter the second number: ");
   scanf("%d", &b);
   printf("\n----- MENU -----\n");
   printf("1. Bitwise AND (&)\n");
   printf("2. Bitwise OR (|)\n");
   printf("3. Bitwise XOR (^)\n");
   printf("4. Bitwise NOT (~)\n");
   printf("5. Left Shift (<<)\n");
   printf("6. Right Shift (>>)\n");
   printf("\nEnter your choice: ");
   scanf("%d", &choice);
   switch(choice)
   {
      case 1:
         res = a & b;
         printf("Bitwise AND Result = %d", res);
         break;
      case 2:
         res = a | b;
         printf("Bitwise OR Result = %d", res);
         break;
```

```
        case 3:
            res = a ^ b;
            printf("Bitwise XOR Result = %d", res);
            break;
        case 4:
            res = ~a;
            printf("Bitwise NOT Result = %d", res);
            break;
        case 5:
            res = a << b;
            printf("Left Shift Result = %d", res);
            break;
        case 6:
            res = a >> b;
            printf("Right Shift Result = %d", res);
            break;
        default:
            printf("Invalid choice.");
    }
    return 0;
}
```

OUTPUT:

```
===== BITWISE OPERATIONS =====
Enter the first number: 12
Enter the second number: 5
----- MENU -----
1. Bitwise AND (&)
2. Bitwise OR (|)
3. Bitwise XOR (^)
```

4. Bitwise NOT (~)

5. Left Shift (<<)

6. Right Shift (>>)

Enter your choice: 1

Bitwise AND Result = 4